

Lab 3: Relational Algebra

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1. Relational Algebra Queries

Query 1

Find the first and last names of persons who have made at least one equipment reservation.

$$\begin{aligned} & \rho(\text{Makers}, \pi_{\text{madeByPid}}(\text{Reservation})) \\ & \pi_{\text{firstName}, \text{lastName}}(\text{Makers} \\ & \quad \bowtie_{\text{Makers.madeByPid}=\text{Person.pid}} \text{Person}) \end{aligned}$$

Explanation: I take the persons who appear in Reservation, then I join with Person to get their names.

Query 2

Find the IDs of persons who are either attached to a laboratory or act as PI of at least one research project.

$$\begin{aligned} & \rho(\text{InLab}, \pi_{\text{pid}}(\text{Attached})) \\ & \rho(\text{PIs}, \pi_{\text{pid}}(\sigma_{\text{role}='PI'}(\text{Participates}))) \\ & \text{InLab} \cup \text{PIs} \end{aligned}$$

Explanation: “Either . . . or” is a union of the two sets of person IDs.

Query 3

Find the IDs of laboratories located in the building ‘College of Computing’ or containing at least one equipment unit whose model belongs to the category ‘CPU Server’.

$$\begin{aligned} & \rho(\text{CPULabs}, \pi_{\text{labId}}(\sigma_{\text{category}='CPU Server'}(\text{EquipmentModel})) \\ & \quad \bowtie_{\text{EquipmentModel.modelId}=\text{EquipmentUnit.modelId}} \text{EquipmentUnit})) \\ & \rho(\text{CoCLabs}, \pi_{\text{labId}}(\sigma_{\text{building}='College of Computing'}(\text{Laboratory}))) \\ & \text{CPULabs} \cup \text{CoCLabs} \end{aligned}$$

Explanation: The two conditions use different tables, so I build one set of labs for each condition and take the union.

Query 4

Find the IDs of laboratories that contain at least one equipment unit whose model belongs to the category ‘CPU Server’ and at least one whose model belongs to the category ‘GPU Server’.

$$\begin{aligned} & \rho(\text{CPULabs}, \pi_{\text{labId}}(\sigma_{\text{category}='CPU Server'}(\text{EquipmentModel})) \\ & \quad \bowtie_{\text{EquipmentModel.modelId}=\text{EquipmentUnit.modelId}} \text{EquipmentUnit})) \\ & \rho(\text{GPULabs}, \pi_{\text{labId}}(\sigma_{\text{category}='GPU Server'}(\text{EquipmentModel})) \\ & \quad \bowtie_{\text{EquipmentModel.modelId}=\text{EquipmentUnit.modelId}} \text{EquipmentUnit})) \\ & \text{CPULabs} \cap \text{GPULabs} \end{aligned}$$

Explanation: σ looks at one row at a time, and one row cannot be CPU and GPU together. So I build one set of labs for CPU, one for GPU, and take the intersection.

Query 5

Find the Faculty members who supervise every laboratory located in the building ‘College of Computing’.

$$\begin{aligned} & \rho(A, \pi_{supervisorPid, labId}(Laboratory)) \\ & \rho(B, \pi_{labId}(\sigma_{building='College of Computing'}(Laboratory))) \\ & Faculty \bowtie_{Faculty.pid=supervisorPid} (A/B) \end{aligned}$$

Explanation: “Every” is a division. A has the pairs (supervisor, lab). B has the labs of the College of Computing. A/B gives the supervisors that are paired with all labs of B . Then I join with Faculty.

Query 6

Find the IDs of persons who participate in every research project funded by at least one budget that also funds the laboratory with labId = 2.

$$\begin{aligned} & \rho(A, \pi_{pid, projectCode}(Participates)) \\ & \rho(LabBudgets, \pi_{budgetLine}(\sigma_{labId=2}(FundsLab))) \\ & \rho(B, \pi_{projectCode}(LabBudgets \\ & \quad \bowtie_{LabBudgets.budgetLine=FundsPrj.budgetLine} FundsPrj)) \\ & A/B \end{aligned}$$

Explanation: B is the set of projects that have a budget line in common with lab 2. “Every” is again a division: A/B gives the persons paired with all projects of B .

Query 7

Find pairs of persons (p_1, p_2) such that p_1 has made more reservations than p_2 .

$$\begin{aligned} & \rho(C, \gamma_{madeByPid}(COUNT(resId) \rightarrow n)(Reservation)) \\ & \rho(C_1(1 \rightarrow p_1, 2 \rightarrow n_1), C) \\ & \rho(C_2(1 \rightarrow p_2, 2 \rightarrow n_2), C) \\ & \rho(Res(1 \rightarrow pid), \pi_{madeByPid}(Reservation)) \\ & \rho(Zero(1 \rightarrow p_2), \pi_{pid}(Person) - Res) \\ & \pi_{p_1, p_2}(C_1 \bowtie_{n_1 > n_2} C_2) \cup (\pi_{p_1}(C_1) \times Zero) \end{aligned}$$

Explanation: C gives the number of reservations of each person. I use C twice (C_1 and C_2) and join with $n_1 > n_2$. A person with no reservation is not in C , because γ makes no group for zero rows. So I take all persons minus the persons in Reservation ($Zero$). Each person of C_1 has at least one reservation, so each one has more than every person of $Zero$. The cross product adds these pairs.

Query 8

Find the IDs of persons who have reserved equipment units located in at least two different laboratories.

$$\begin{aligned} & \rho(R, \pi_{madeByPid, labId}(Reservation \\ & \quad \bowtie_{Reservation.resId=Reserves.resId} Reserves \\ & \quad \bowtie_{Reserves.serialNo=EquipmentUnit.serialNo} EquipmentUnit)) \\ & \rho(G, \gamma_{madeByPid}(COUNT(labId) \rightarrow k)(R)) \\ & \pi_{madeByPid}(\sigma_{k \geq 2}(G)) \end{aligned}$$

Explanation: π removes duplicate (person, lab) pairs, so $COUNT$ counts different labs. Then σ keeps $k \geq 2$.

Query 9

Find the IDs of reservations submitted in September 2026 that reserve at least one equipment unit currently located in the laboratory named 'AI Movement'.

$$\begin{aligned}
& \rho(\text{Sep}, \sigma_{\text{submissionTS} \geq '2026-09-01' \wedge \text{submissionTS} < '2026-10-01'}(\text{Reservation})) \\
& \rho(\text{AI}, \sigma_{\text{name}='AI Movement'}(\text{Laboratory})) \\
& \pi_{\text{Sep.resId}}(\text{Sep} \\
& \quad \bowtie_{\text{Sep.resId}=\text{Reserves.resId}} \text{Reserves} \\
& \quad \bowtie_{\text{Reserves.serialNo}=\text{EquipmentUnit.serialNo}} \text{EquipmentUnit} \\
& \quad \bowtie_{\text{EquipmentUnit.labId}=\text{AI.labId}} \text{AI})
\end{aligned}$$

Explanation: *Sep* keeps the reservations from September 1 (included) to October 1 (excluded). *AI* is the lab named 'AI Movement'. The joins connect the reservation to its units and the units to the lab.

Query 10

Find the IDs of persons who act as PI of more than one research project.

$$\begin{aligned}
& \rho(P, \pi_{\text{pid}, \text{projectCode}}(\sigma_{\text{role}='PI'}(\text{Participates}))) \\
& \rho(G, \gamma_{\text{pid}}(\text{COUNT}(\text{projectCode}) \rightarrow n)(P)) \\
& \pi_{\text{pid}}(\sigma_{n>1}(G))
\end{aligned}$$

Explanation: I keep the PI rows, count the projects of each person with γ , then keep $n > 1$.

Query 11

Find the IDs of approvers who have approved reservations involving equipment units located in more than one laboratory.

$$\begin{aligned}
& \rho(R, \pi_{\text{approverPid}, \text{labId}}(\sigma_{\text{status}='Approved' \vee \text{status}='Completed'}(\text{Reservation})) \\
& \quad \bowtie_{\text{Reservation.resId}=\text{Reserves.resId}} \text{Reserves} \\
& \quad \bowtie_{\text{Reserves.serialNo}=\text{EquipmentUnit.serialNo}} \text{EquipmentUnit})) \\
& \rho(G, \gamma_{\text{approverPid}}(\text{COUNT}(\text{labId}) \rightarrow k)(R)) \\
& \pi_{\text{approverPid}}(\sigma_{k>1}(G))
\end{aligned}$$

Explanation: Same pattern as Query 8, but for the approver. I keep approved reservations, join to the units to get the labs, remove duplicates with π , count labs per approver, and keep $k > 1$.

Query 12

Find the IDs of persons who have no reservation whose planned reservation interval includes November 6, 2026.

$$\begin{aligned}
& \rho(\text{Busy}(1 \rightarrow \text{pid}), \pi_{\text{madeByPid}}(\\
& \quad \sigma_{\text{plannedStart} < '2026-11-07' \wedge \text{plannedEnd} \geq '2026-11-06'}(\text{Reservation}))) \\
& \pi_{\text{pid}}(\text{Person}) - \text{Busy}
\end{aligned}$$

Explanation: A reservation includes November 6 if it starts before November 7 and ends on or after November 6. I find the persons who have such a reservation and subtract them from all persons.

Query 13

Find the laboratories whose number of equipment units is below the average number of equipment units per laboratory.

$$\begin{aligned}
& \rho(C, \gamma_{labId}(COUNT(serialNo) \rightarrow n)(EquipmentUnit)) \\
& \rho(T, \gamma_{COUNT(serialNo) \rightarrow s}(EquipmentUnit)) \\
& \rho(L, \gamma_{COUNT(labId) \rightarrow c}(Laboratory)) \\
& \rho(AvgU, \pi_{s/c \rightarrow a}(T \times L)) \\
& \rho(Low_1, \pi_{labId}(\sigma_{n < a}(C \times AvgU))) \\
& \rho(Zero, \pi_{labId}(Laboratory) - \pi_{labId}(EquipmentUnit)) \\
& \rho(Low, Low_1 \cup Zero) \\
& Laboratory \bowtie_{Laboratory.labId=Low.labId} Low
\end{aligned}$$

Explanation: C gives the number of units n of each lab. T gives the total number of units s . L gives the number of laboratories c . The average is $a = s/c$. σ compares columns of the same row, so I attach a to every row of C with a cross product, then keep $n < a$. A lab with no unit is not in C , but its number of units is 0, which is below the average. So I add these labs with $Zero$ (all labs minus the labs in $EquipmentUnit$).

Query 14

For each laboratory, return the person who has made the greatest number of distinct approved reservations involving equipment units located in that laboratory.

$$\begin{aligned}
& \rho(R, \pi_{labId, madeByPid, resId}(\sigma_{status='Approved' \vee status='Completed'}(Reservation)) \\
& \quad \bowtie_{Reservation.resId=Reserves.resId} Reserves \\
& \quad \bowtie_{Reserves.serialNo=EquipmentUnit.serialNo} EquipmentUnit)) \\
& \rho(C, \gamma_{labId, madeByPid}(COUNT(resId) \rightarrow n)(R)) \\
& \rho(M(1 \rightarrow labM, 2 \rightarrow m), \gamma_{labId}(MAX(n) \rightarrow m)(C)) \\
& \rho(Best, \pi_{labId, madeByPid}(C \\
& \quad \bowtie_{C.labId=M.labM \wedge C.n=M.m} M)) \\
& Best \bowtie_{Best.madeByPid=Person.pid} Person
\end{aligned}$$

Explanation: π keeps (lab, person, reservation) once, so a reservation with several units in the same lab is counted once. C counts the reservations of each person in each lab. M gives the maximum count of each lab. The join with M keeps the persons whose count is equal to the maximum. The last join gives the person information.

Query 15

List the IDs of persons who made at least three equipment reservations during the year 2025.

$$\begin{aligned}
& \rho(R, \sigma_{submissionTS \geq '2025-01-01' \wedge submissionTS < '2026-01-01'}(Reservation)) \\
& \rho(G, \gamma_{madeByPid}(COUNT(resId) \rightarrow n)(R)) \\
& \pi_{madeByPid}(\sigma_{n \geq 3}(G))
\end{aligned}$$

Explanation: I keep the reservations of 2025, count them per person with γ , then keep $n \geq 3$.

2. Challenges

Getting started

At first I did not know how to turn a sentence into an algebra expression, and I also had to learn how to write it in L^AT_EX. Each query took a long time, because I kept going back to the Lab 2 solution to

check the exact names of tables and attributes, and sometimes to Lab 1 to understand the assumptions. For some queries, I created small example tables on paper and checked the result by hand.

Joins

I first wanted to use a natural join, but `madeByPid` and `pid` have different names, so nothing matches. I wrote every join condition explicitly. For Query 7 I needed the same relation twice, so I used ρ to rename the columns.

Union, intersection, and, or

Queries 3 and 4 look the same, but they are not. In Query 4, putting `CPU Server` $\wedge$ `GPU Server` in one σ returns nothing, because a unit has only one category. Putting $\vee$ returns laboratories with only one of the two. σ checks one row at a time, so I built one set of laboratories for CPU and one for GPU, and used $\cap$.

Every and no

For “every” (Queries 5 and 6) I used division. The difficult part was choosing A and B . For Query 12 “no reservation” cannot be done with σ , because σ cannot find persons who have no matching row. So I computed the persons who have a reservation on November 6 and removed them from all persons.

Counting and average

Aggregation was new for me. σ looks at one row at a time, and γ works on groups of rows. For “at least three” I use γ first, then σ . In Query 13 the average has to be computed inside the query, and since σ compares columns of the same row, I attach it to every laboratory with a cross product. I divide the total number of units by the number of laboratories, so laboratories without units are included in the average.

Groups with zero rows

γ does not create a group for a person or a laboratory with zero rows. In Query 7, a person with no reservation would never appear as p_2 , so I add the persons with no reservation (all persons minus the persons in Reservation). In Query 13, a laboratory with no unit would never appear in C , so I add the laboratories with no unit (all laboratories minus the laboratories in EquipmentUnit).

3. Assumptions

- The schema keeps only the current laboratory of a unit. So in Queries 8, 9, 11 and 14, the laboratory of a unit is *EquipmentUnit.labId*.
- A reservation is approved if its status is Approved or Completed (Queries 11 and 14).
- “Submitted in” and “made during” use the column `submissionTS` (Queries 9 and 15).
- In Query 11, an approver is returned if the laboratories of all their approved reservations together are more than one.
- In Query 14, if several persons have the same maximum in a laboratory, all of them are returned. A laboratory with no approved reservation returns nothing.
- In Query 13, I assume at least one unit exists.